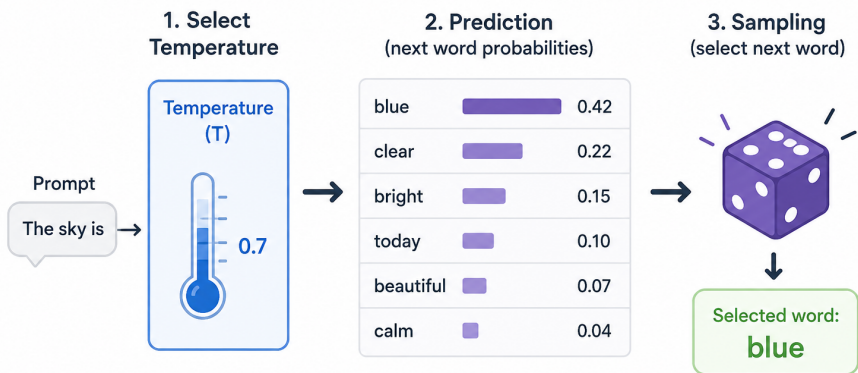




CORE CONCEPTS

Temperature & Sampling in LLMs

The knobs that control how your LLM thinks and why most people set them wrong.



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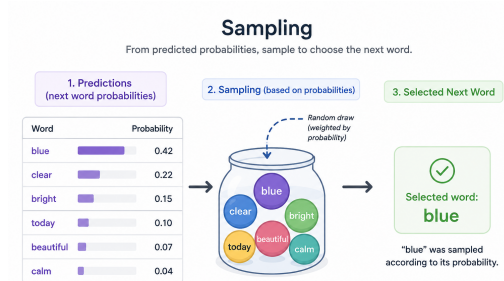
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What is Sampling & Why It Matters

Every time an LLM generates a token, it's making a probabilistic choice not a deterministic one.

✍ The model doesn't "write" text. It assigns probability scores to every possible next token, then picks one.

⚙ How it picks that token is controlled by sampling parameters.



⚡ Real Scenario

You're building a legal document summarizer.

Wrong settings on the legal tool = hallucinated clauses.

A colleague uses the same model for a creative story generator.

Wrong settings on the story tool = robotic output.

⚠ Same model. Completely different sampling settings. Completely different behavior.



How Token Probability Works

Before explaining the parameters you need to understand what they operate on.

SCENARIO

Model is mid-sentence: "The capital of France is ____"

"Paris"	94%
"Lyon"	2%
"France"	1.5%
"a"	0.8%

→ Without any sampling model always picks "Paris" (argmax / greedy decoding).

↔ With sampling the model can pick from the distribution, introducing variety.

WHY GREEDY DECODING FAILS

Always picking the highest probability token leads to repetitive, generic output:
"The best way to learn is to learn by doing and learning from your mistakes..."



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Temperature

What it controls: How sharp or flat the probability distribution is before sampling.

- ↓ **Low temp (0.0 - 0.3)** sharpens the distribution, makes the top token dominate.
- ↑ **High temp (0.8 - 2.0)** flattens the distribution, gives lower tokens more chance.

Low temperature (0.1)

You ask: "What is the boiling point of water?"

Model is near-certain. "100°C" dominates at 99%+. Low temp keeps it deterministic. Right call.

High temperature (1.2)

You ask: "Write me a tagline for a coffee brand"

High temp explores more of the distribution. More creative variance. Right call.

WHEN TO TUNE

0.0 - 0.3: Factual QA, RAG, legal, medical

0.5 - 0.7: Balanced assistants, chat

0.8 - 1.2: Creative writing, marketing copy



Top-k Sampling

What it controls: The number of candidate tokens the model considers at each step.

- ✓ **Top-k = 1** always pick the highest probability token (greedy).
- ≡ **Top-k = 50** only consider the top 50 tokens. Everything else is cut to zero.

☰ "SHE OPENED THE DOOR AND ____"

"walked" 35%

"entered" 25%

"saw" 18%

"found" 12%

"slammed" 6%

"danced" 1.5%

Top-k = 3 only "walked", "entered", "saw" are candidates.

Top-k = 50 includes "danced" and other unlikely picks.

⚠ THE PROBLEM WITH TOP-K

k is a fixed number. Sometimes top 50 are all reasonable, sometimes only top 2 make sense. It doesn't adapt to how "certain" the model is.



Top-p (Nucleus)

What it controls: Consider tokens until their cumulative probability reaches p .

⚙️ THE NUCLEUS

Top-p = 0.9 keep adding tokens from highest to lowest probability until the running total hits 90%. It dynamically sizes the candidate pool.

🔒 High Certainty

"The capital of France is ____"
"Paris" is 94%. At top-p = 0.9, only
"Paris" is in the nucleus. Correct
behavior.

🔓 Low Certainty

"She felt ____"
"happy" (18%), "sad" (15%),
"nervous" (12%)...
At top-p = 0.9, many tokens are
needed to reach 90%. Wide
nucleus. More creative picks.

🏆 WHY TOP-P BEATS TOP-K

It adapts to the model's confidence automatically. Same parameter,
intelligent behavior in both cases.



How They **Interact**

These parameters don't work in isolation they stack.

- 📦 **Temperature** reshapes the distribution first.
- 📦 **Top-k** then cuts it to the top k tokens.
- 📦 **Top-p** then cuts further to the nucleus.

❌ **BAD CONFIG**

```
temp=1.5, top_p=0.95,  
top_k=500
```

Too much randomness. Bot adds creative speculation to factual answers.

✅ **GOOD CONFIG**

```
temp=0.2, top_p=0.5,  
top_k=20
```

Tight, deterministic, grounded.
Same query = same accurate answer.

💡 **Common Mistake:** Setting high temp AND high top-p AND high top-k together compounds into maximum randomness. Pick one lever to push high.



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Real-World Recipes

Legal Summarizer

Accurate, consistent, zero hallucination.

T: 0.1 | P: 0.3 | K: 10

Support Chatbot

Helpful & natural, but factually grounded.

T: 0.4 | P: 0.7 | K: 50

Marketing Copy

Creative, punchy, varied outputs.

T: 0.9 | P: 0.92 | K: 200

Code Assistant

Correct & consistent. Creativity is a bug.

T: 0.1 | P: 0.2 | K: 5

Creative Story Writer

Maximum variety, unexpected turns, rich language.

T: 1.2 | P: 0.95 | K: 500



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Parameter Cheatsheet

Parameter	Low Value	High Value	Use Case	Temp	Top-p	Top-k
Temperature Distribution	Focused	Creative	RAG / QA	0.1	0.3	10
Top-k Fixed count	Few options	Many options	Support	0.4	0.7	50
Top-p Mass cutoff	Tight nucleus	Wide nucleus	Code Gen	0.1	0.2	5
			Marketing	0.9	0.92	200
			Creative	1.2	0.95	500

★ GOLDEN RULES

- When correctness matters turn down all three.
- When creativity matters temp is your primary lever, top-p secondary.
- Never max out all three simultaneously.



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